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Review

Empowering natural human–robot collaboration through multimodal language models and spatial intelligence: Pathways and perspectives

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ABSTRACT

Industry 5.0 advocates human-centric smart manufacturing (HSM), with growing attention to proactive human-machine collaboration (HRC). Meanwhile, the rapid development of Multimodal large language models (MLLMs) and embodied intelligence is driving an unprecedented evolution. This work aims to leverage these opportunities to enhance robots' learning and cognitive capabilities, enabling seamless and natural interaction. However, current research often overlooks human–robot symbiosis and lacks attention to specialized models and practical applications. This review adheres to a human-centric vision, taking language as the pivot to connect humans with large models. To our best knowledge, this is the first attempt to integrate HRC, MLLMs and embodied intelligence into a holistic view. The review first introduces representative foundation models to provide a comprehensive summary of state-of-the-art methods in the "Perception-Cognition-Actuation" loop. It then discusses pathways and platforms for efficient spatial skills learning, followed by an analysis of four key questions from the "Why, How, What, Where" perspectives. Finally, it highlights future challenges and potential research directions. It is hoped that this work can help fill the research gap between HRC and MLLMs, offering a systematic pathway for developing human-centered collaborative systems and promoting further exploration and innovation in this exciting and crucial field. The resources are available at: https://github.com/WuDuidi/MLLM-HRC-Survey.

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1. Introduction

In recent years, robot learning has undergone multi-stage evolutionary development. The first stage involved robot learning through teaching programming, followed by interaction-based reinforcement learning. The third stage is embodied imitation learning, and the most recent focus has been on learning models based on Artificial Intelligence-Generated Content (AIGC). Meanwhile, the human–robot relationship has also progressed through five stages, referred to as the "5C": Coexistence, Cooperation, Collaboration, Compassion, and Coevolution. These stages range from "Coexistence", where tasks are alternately executed in shared workspaces, and ultimately to "Collaboration", where tasks are performed simultaneously with potential direct interaction [1]. This progression marks a shift toward deeper interaction and symbiosis. The advancement of Artificial Intelligence and Industry 5.0 further drive this trend, emphasizing Human-centric Smart Manufacturing (HSM) and Human–Cyber–Physical Systems (HCPS). This has sparked widespread interest and exploration in Human-Robot Collaboration (HRC) [2], the paradigm that leverages human cognitive flexibility alongside robotic automation capabilities to achieve higher production efficiency. In this context, intelligent agents' adaptive capabilities become crucial. Within open environments, agents must comprehend and accommodate human intentions, explore learning objectives, and align with human values to facilitate more effective collaboration and autonomous learning, ultimately enhancing human well-being.

The real manufacturing world is a dynamic, open, multimodal, multi-party involved environment. Intelligent agents must process information from different modalities, adapt to unknown scenarios, and reach higher levels of intelligence through interactions with other agents and humans. Early multimodal fusion for human–robot interaction primarily focused on several modes: visual, auditory, tactile, and physiological perceptions (EEG, EMG, ECoG etc.) [3,4].

In addition to multi-sensor signals, natural language serves as a key medium for transferring intelligence. Humans exhibit complex intelligent behaviors through advanced expression and abstraction capabilities. The field of Natural Language Processing (NLP) has witnessed the rise of Pre-trained Language Models, developed on Transformer architectures and trained on massive text datasets. As these models scale to billions or even trillions of parameters, the so-called Large Language Models (LLMs) have achieved groundbreaking performance. The remarkable success of LLMs has sparked exploration into their integration with multimodal systems, with the emergence of Vision-Language Models (VLMs) and Vision-Language-Action Models (VLAs), which provide a pathway from perception to action, further advancing the development of Multimodal Large Language Models (MLLMs). Simultaneously,

Nomenclature	
AIGC	Artificial Intelligence-Generated Content
AGI	Artificial General Intelligence
HSM	Human-centric Smart Manufacturing
HCPS	Human–Cyber–Physical Systems
HRC	Human-Robot Collaboration
HRI	Human-Robot Interaction
HCI	Human-Computer Interaction
CV	Computer Vision
NLP	Natural Language Processing
MLLM	Multimodal Large Language Model
LLM	Large Language Model
VLP	Visual-Language Pretrained Model
VFM	Vision-based Foundation Model
VLM	Vision-Language Model
VLA	Vision-Language-Action Model
OOD	Out-of-Distribution
CoT	Chain of Thought
VQA	Visual Question Answering
EEG	Electroencephalography
BCI	Brain-Computer Interface
XR	Extended Reality

this advancement fosters the paradigm of "Embodied Intelligence", a concept first proposed by Alan Turing in the early 1950s, which embeds AI into physical entities for dynamic environmental interaction, thereby extending AI's capabilities into the physical world.

As shown in Fig. 1, these developments in computer intelligence such as computer vision and generative reasoning, have provided smarter pathways for human–robot interaction, endowing robots with the abilities to "see" and "think". Moreover, embodied intelligence uses AI as the "brain" and integrates grounded physical motor systems as the "cerebellum", bridging human–computer interaction (HCI) with human–robot interaction (HRI). The evolving relationship between humans and robots reveals two key dimensions for achieving the ability to "do":

- **Human-Robot Collaboration**: Robots learn human operational and decision-making patterns through skill transfer to achieve enhanced collaborative intelligence.
- **Human-Robot Symbiosis**: Robots develop autonomous skills and interact physically with their environment, exemplified by humanoid robots, brain–computer interfaces, and wearable devices, fostering a deep harmonious relationship.

Indeed, these two dimensions are cyclically complementary, with embodied robots driving enhanced intelligence, while enhanced intelligence refines embodied cognitive abilities, ultimately working together to better assist humans.

Significantly, an essential point to emphasize is that achieving truly proactive human–robot collaboration [5] requires robots to possess advanced multimodal natural interaction capabilities and spatially operable intelligence, referred to as **"Spatial Intelligence".** This is critical for understanding and interacting with the 3D world, with its core being the construction of a "world model" that endows robots with human-like perceptual awareness. As depicted in Fig. 1, a key pathway to achieving spatial intelligence lies in digitalizing the physical world and exploring robotic skill acquisition in cyberspace, thereby enabling iterative experimentation in simulations and enhancing real-world intelligence through sim-to-real transfer.

1.1. Research objectives and scope

Enhancing the learning and cognitive capabilities of robots to achieve proactive, seamless and natural human–robot interaction has been a long-term goal. However, robotic skills (e.g., workflows, trajectories) are typically pre-programmed for fixed workstations in existing manufacturing, where HRC struggles in unstructured environments or with unpredictable interventions, leading to inefficiencies and limitations in customized tasks. To address these challenges, the objectives of this work focus on two key aspects: **(1) Few-shot and rapid learning**. Integrating multimodal data positions language as a crucial bridge between natural human–robot interaction and large models, with MLLMs' powerful reasoning and generalization capabilities offering a promising pathway for swift adaptation to specific tasks. **(2) Low-cost and multi-skill systems**. It is worth noting that physical and spatial interactions remain crucial considerations, as the real world demands strict error-free operations to ensure safety. In particular, spatial intelligence provides a valuable solution by enabling efficient, safe, and cost-effective learning in cyberspace while simultaneously learning multiple skills. These two approaches represent the core outlined in the title.

1.2. Research gap and questions

Several reviews have explored the integration of foundation models with robotics, but most are outdated, particularly those published before 2024 [9–11]. Although two updates released in Oct.2024 [12, 13], this field is experiencing unprecedented rapid evolution catalyzed by the waves of Embodied AI and Agent AI [14]. However, these existing works advocate for generalization in desktop and daily operations, while lacking consideration of human factors and exploration in human–robot collaboration scenarios, also with limited attention to task-specific models for manufacturing applications.

To bridge this gap, guided by a human-centric vision, this work conducts a systematic review aimed at addressing the following questions.

1. ***RQ1**: How to leverage emerging MLLMs and embodied intelligence to enhance human-like AI and achieve seamless human–robot interaction?*
2. ***RQ2**: How to train and learn spatial skills efficiently for specific models?*
3. ***RQ3**: What are the future challenges and potential directions?*

The rest of this article is organized as shown in Fig. 2: Section 3 provides a comprehensive introduction to existing state-of-the-art approaches within the "Perception-Cognition-Actuation" loop. Section 4 summarizes the pathway for specialist agents and platforms toward spatial intelligence. Section 5 delves into four critical questions through the dimensions of "Why, How, What, and Where". Section 6 discusses promising research directions, while Section 7 offers concluding remarks. This work serves as the first attempt to integrate HRI, embodied intelligence, and spatial intelligence into a holistic view. It is hoped that this work can help to fill the research gap at the intersection of HRC and MLLMs and provide a systematic perspective to advance human-centered collaborative systems.

2. Preliminaries

2.1. Overview of foundation models evolution

By nature, the realization of human-like AI lies in unified models capable of addressing diverse real-world challenges, while traditional models trained on domain-specific data often struggle with versatility. Thus, the trend shifts toward pre-trained systems adaptable to multiple tasks and environments. Accordingly, foundation models represent a significant step to address this challenge. Foundation models are advanced architectures trained on large-scale, diverse datasets, capable of adapting to a wide range of downstream tasks through fine-tuning or zero-shot learning. Pre-trained foundation models [16] have paved the way for the rapid development of multimodal large models. The successful deployment of models like OpenAI's GPT-4V [17] and Google's Gemini [18] showcased remarkable multimodal capabilities, progression from LLMs to VLPs, and finally to MLLMs.

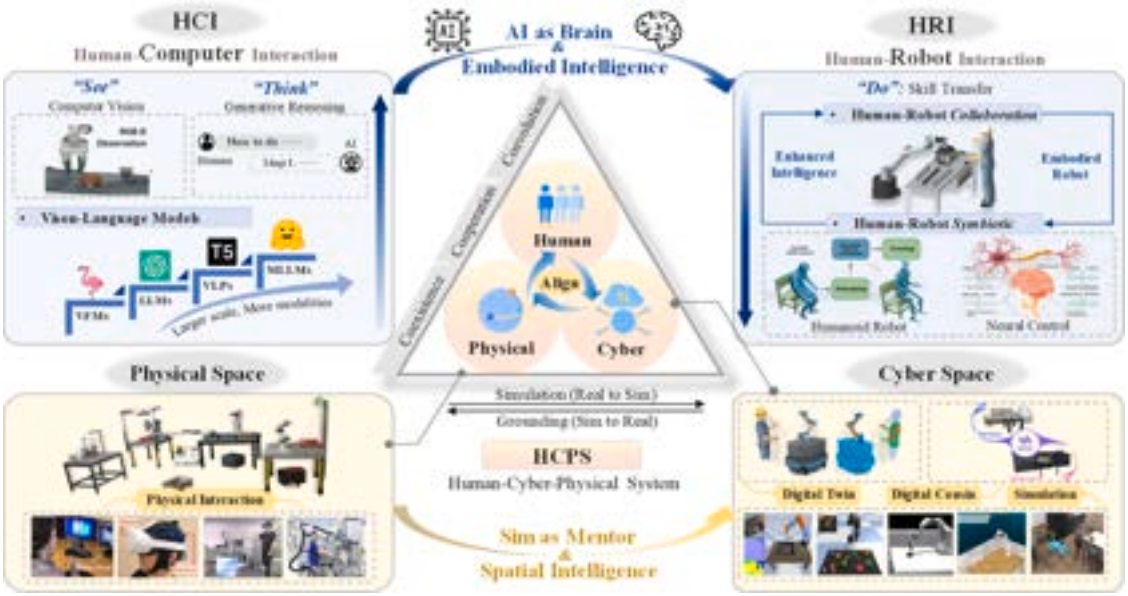

Fig. 1. Human-Cyber-Physical space alignment with embodied intelligence and spatial intelligence promotes human–computer interaction toward human–robot symbiosis. *Source:* Adapted elements from [5–8].

Fig. 2. The overall structure of this survey.

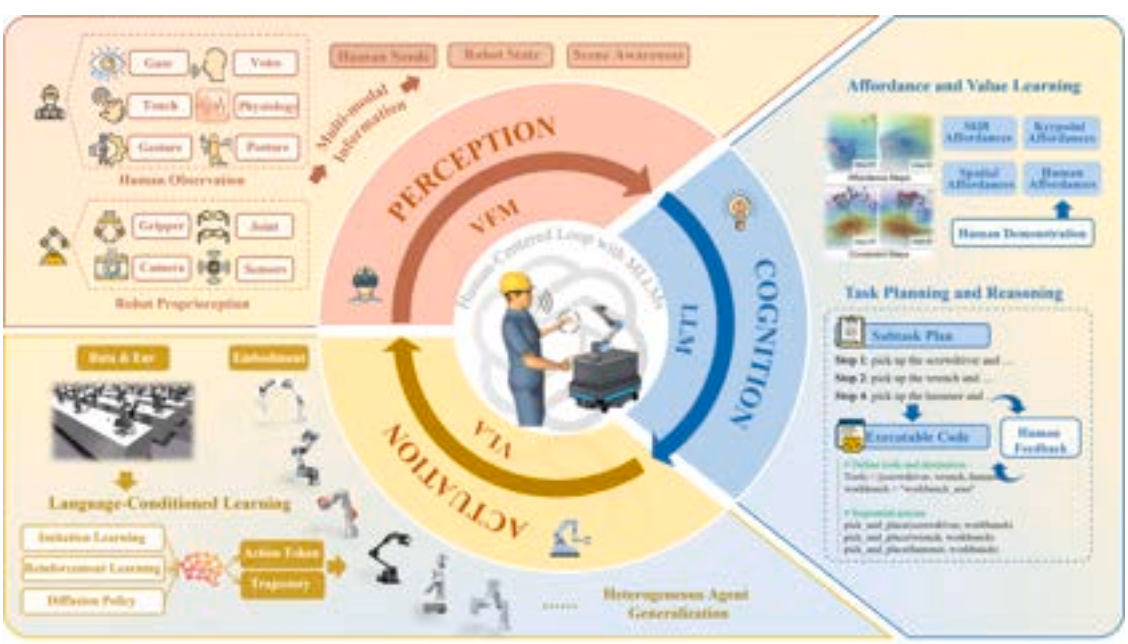

Fig. 3. Human-centered loop with MLLMs for enhanced robot intelligence and seamless interaction. *Source:* Adapted elements from [5,15].

2.1.1. LLMs

LLMs are trained on internet-scale datasets, processing trillions of tokens, and exhibit emergent capabilities such as in-context learning, instruction following, and multi-step reasoning. The representative families are as follows.

The GPT family, developed by OpenAI, is based on autoregressive Transformer architecture. GPT-2 (1.5 billion parameters) gained attention for its performance in text generation. GPT-3 (175 billion parameters) enhanced zero-shot and few-shot learning, while InstructGPT [19] and ChatGPT (based on GPT-3.5) improved instruction following and dialogue generation through Reinforcement Learning with Human Feedback (RLHF). GPT-4 further increased parameter scale and multimodal capabilities.

The LLaMA [20] family, launched by Meta (Facebook AI), includes models such as LLaMA-13B, which outperformed GPT-3 (175B) on benchmark tests despite having fewer parameters. Alpaca [21] fine-tuned LLaMA to enhance instruction following, while Vicuna [22] optimized dialogue generation using human feedback. The PaLM family, developed by Google AI, focuses on multi-task and few-shot learning. PaLM demonstrated exceptional multi-task generalization, while PaLM-E [23] embodied visual-language knowledge for reasoning and supports robot planning.

2.1.2. VFMs

Alongside significant advancements in Natural Language Processing (NLP), Vision-based Foundation Models (VFMs) have also emerged. Vision Transformer (ViT) [24] pioneered this field by adapting Transformers with variants like ViT-L, ViT-B. Enhanced architectures such as DeiT [25] and PVT [26] improved performance through techniques like knowledge distillation. Florence [27] expands representations from coarse (scene) to fine (object) and from static (images) to dynamic (videos). VFMs for detection and segmentation have been widely adopted for visual representation and open-world recognition, as detailed in Section 3.1.2. Additionally, Visual-Language Pre-trained models (VLPs) further leverage visual-language fusion, categorized into contrastive learning models like CLIP and generative models like VL-T5 [28] and Flamingo [29]. These pre-trained models serve as a foundation for fine-tuning, highlighting the broad application and potential in multimodal tasks.

2.1.3. VLAs

Vision-Language-Action Models (VLAs) are one of the core technologies for achieving embodied intelligence, enabling direct translation from language instructions and visual understanding to executable actions. As an extension of LLMs and VLMs, VLAs integrate internet-scale knowledge, physical-world concepts, and motion information into a unified framework. RT-1 [30] first introduced the VLA concept, utilizing a Transformer architecture to combine visual and linguistic inputs for generating discrete action tokens. VIMA [31] demonstrated the ability to learn manipulation tasks from multimodal prompts. RT-2 [32] further advanced this paradigm by integrating VLMs with robot motion data through imitation learning, enabling the generation of actionable robot commands. The versatility of VLAs is exemplified by their application across diverse domains, such as quadruped robots (e.g., Quarl-VLA [33]) and autonomous driving (e.g., CoVLA [34]), highlighting their broad potential and adaptability in robotics and beyond.

2.2. MLLMs: Architectures and advancements

Building on the advancements of the previously discussed foundation models, Multimodal Large Language Models (MLLMs) have emerged as a critical research frontier, driven by the expansion of data and model scale. LLaVA [35] introduced a novel approach through joint embedding spaces and multi-layer attention mechanisms, while the release of GPT-4V dramatically accelerated MLLMs development, showcasing unprecedented multimodal capabilities. These models integrate modalities in diverse ways:

- MM2L (Vision + Language → Latent Space): Enables similarity computation for tasks like visual retrieval [36–38];
- MM2T (Vision + Language → Text): Supports tasks such as image captioning and visual question answering (VQA) [29];
- MM2I (Vision + Language → Image): Utilizes VAE and diffusion models for image generation and editing based on language input [39].
- MM2A (Vision + Language → Action): Translates multimodal inputs into physical actions, enabling embodied intelligence and robotic control.

Recent advancements have significantly expanded MLLMs capabilities, with universal models like AnyGPT [40], NExT-GPT [41], and ModaVerse [42] integrating diverse modalities including audio and video, to enhance flexibility and broader applications, as well as incorporating sensory information such as tactile [43], paving the way toward Artificial General Intelligence (AGI).

3. Integrating MLLMs into the HRC loop

Unlike traditional task-specific models, MLLMs leverage natural language as an interface, bridging the gap between human intent and robotic execution. As shown in Fig. 3, MLLMs play a pivotal role in the human-centered loop. In perception, they enable the understanding of multi-modal information, including human needs, robot states, and scene awareness, with VFMs enabling essential recognition capabilities. In cognition, affordances are learned from human demonstrations, while task reasoning generates plans and executable code, driven primarily by LLMs. In actuation, language-conditioned policies facilitate generalization across heterogeneous embodiments, enabling physical interaction and seamless multi-agent collaboration, with VLAs as the core technology. In this section, cutting-edge advancements and their potential will be investigated, answering ***RQ1***. The timeline of the main algorithms involved is presented in Fig. 4.

3.1. Perception

3.1.1. Human intent and activity recognition

Unlike traditional industrial robots, cobots in HRC need to perceive both the physiological (e.g., gestures and biosignals) and psychological states (e.g., fatigue and stress) of humans in real time to enable proactive planning and error prevention, which is crucial for achieving safer, more empathetic, and natural collaboration [5]. This process involves multimodal perception. Traditional methods usually integrate visual (RGB, skeleton, and depth) and sensor data (physiological signals, accelerometers, infrared, radar, and point clouds) for intention recognition and environment sensing, such as tactile, speech, and hand motion-driven control [44], and EMG and IMU-based collaboration [45].

With the emergence of VLMs, language modalities have gained increasing attention. Pioneering works like ActionCLIP [46] and Text4Vis [47] reframed action recognition innovatively as a video-to-text retrieval challenge. The AToM-Bot [48] framework leverages VLMs to infer human states and environmental observations, generating context-specific solutions. Similarly, the Language-driven Intention Tracking (LIT) [49] integrates large language and vision models to predict human behaviors in robotic-assisted cooking tasks.

Meanwhile, the combination of foundation models and generative models has also opened novel directions. MotionDiffuse [50] first realizes text-based motion diffusion model, MotionCLIP [51] aligns human motions in latent space to guide motion reconstruction, and MotionGPT [52] enables action generation and recognition. These advancements provide new insights into constructing human digital twins (HDT) and achieving 3D human perception.

Fig. 4. Timeline of multimodal vision-language models for advanced robot planning and low-level execution.

3.1.2. Open-vocabulary object detection

Open-vocabulary detection is critical to identifying unknown objects in dynamic, unstructured environments. Detection models are typically categorized into two types: localization-focused models [38,53,54] which generate bounding boxes, and representation-centric models [55–57], primarily serving as encoders.

GLIP [38] and MDETR [53] redefine object detection as phrase grounding, while OWL-ViT [54] introduced unprecedented flexibility with multi-query interactive retrieval, excelling in zero-shot generalization across robotic perception tasks, including mask generation for target objects [58], bounding box detection [15,59], multimodal interaction [60], and navigation in novel scenes [61]. Visual representation models like DINO [55] combine knowledge distillation and self-supervised learning to equip robust classification capabilities. Grounding DINO [56] further enhances cross-modal fusion and open-set detection. DINO v2 [62] and SigLIP [57] serve as reliable semantic visual encoders for systems like VLA and world models.

Beyond detection, open vocabulary segmentation and scene segmentation are also pivotal for robotic perception. SAM [63] has enabled advanced applications including action mask generation [64] and constraint perception [65]. Its extension, GroundedSAM [66], supports keypoint annotation [67]. Furthermore, increasing studies are combining detection and segmentation, examples include DINOv2+SAM [65,68], OWL-ViT+SAM [15], and DINOv2+Grounded-SAM [69]. These advances significantly enhance semantic and spatial comprehension, laying the foundation for adaptive interactions.

3.1.3. Scene understanding and 3D awareness

However, human-level and object-level perception often fail to address ambiguities in dynamic, complex tasks like off-visual-line navigation, which highlights the need for scene-level perception. The emergence of 3D perception-oriented MLLMs provides new solutions for these challenges.

Point-Bind [70] achieves any-to-3D generation and zero-shot 3D understanding by aligning point clouds with images and language. It evolved through integration with LLaMA into PointLLM [71], introducing capabilities for colored point cloud understanding and contextual generation. SpatialVLM [72] equips VLMs with quantitative spatial reasoning to estimate distances from 2D images.

These developments have catalyzed the evolution of Vision-Language Navigation models (VLN). Such as constructing semantic maps and providing spatial memory for mobile robots through foundation models. Furthermore, LM-Nav [73] achieves long-distance navigation without fine-tuning by leveraging pre-trained models. NLMap [74] supports complex task execution through LLM-based planning. Galbot's NaVid [75] model generates robot operations using only RGB video input, bridging the gap between out-of-distribution (OOD) scenarios and sim-to-real transfer, thus further enhancing generalization capabilities.

3.2. Cognition

Task cognition focuses on active learning from perception results within the "human–robot-task" framework and embedding decision into continuous bidirectional collaboration. Fig. 5 shows robots' cognition of humans. While traditional approaches such as knowledge graphs and decision trees have been widely implemented, the multi-level cognitive capability of foundation models endows robots with greater autonomy and flexibility.

3.2.1. Affordance and value learning

A key challenge in robot planning and decision-making is breaking down high-level instructions and tasks into executable actions. Value functions associate semantic information with the physical environment through multimodal value learning or 3D value maps. Additionally, affordance prediction helps robots understand interaction possibilities, bridging the gap between perception and execution while mitigating language ambiguity. Affordances can be represented through heatmaps [85–87], spatial constraints [65,68], or keypoint constraints [88]. Numerous studies have demonstrated the potential of MLLMs in various applications, Table 1 provides a summary.

Skill affordances (e.g., grasping and placing) are fundamental to robot operations. CLIPort [79] combines CLIP's semantic understanding ("What") with Transporter's spatial precision ("Where") in a dual-stream architecture to handle diverse language-specified tasks. SayCan [80] leverages LLMs ("Say") to set task goals and affordance functions ("Can") to support physical operations. Voltron [81] predicts dense segmentation masks to identify graspable positions, while CoPa [65] generates task-aware 6-DoF grasp planning using VLMs.

Keypoint affordances focus on fine-grained action. MOKA [67] formulates affordance reasoning as a visual question-answering problem to annotate key points in open-world scenarios. KALIE [82] extends this by learning from imagined environments. ReKep [68] represents manipulation tasks as relational key point constraints and predicts trajectory through hierarchical optimization.

Spatial affordances combine navigation with action execution. RAM [69] transfers affordances in 2D and elevates them to 3D contact points and directions to enhance manipulation capabilities. Similarly, RoboPoint [83] predicts operational points and then projects them into 3D for navigation. VoxPoser [15] constructs 3D value maps to map affordances and constraints for spacial trajectory synthesis.

Human affordances advocate using human video for learning. Such as achieving pretrained representations of "affordance labels" for hands, objects, and contacts [84], and extracting human affordances from unstructured data [85]. This paradigm emphasizes the potential of using rich real-world data for more dynamic modeling, and has gained increasing attention as detailed in Section 3.2.3.

3.2.2. Task planning and reasoning

Leveraging large language models, robots can achieve advanced task planning through 3 main approaches: VQA, subtask decomposition, and code generation. Where the VQA capabilities of MLLMs

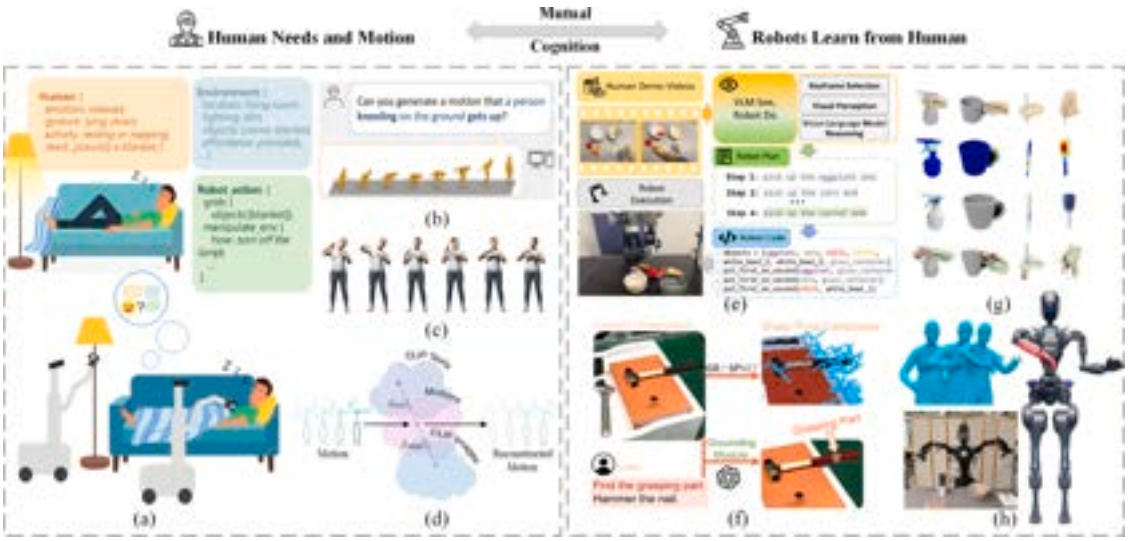

Fig. 5. Mutual cognition between human and robot, covering human intentions ((a) AToM-Bot [48]) and movements perception ((b) MotionGPT [52], (c) MotionDiffuse [50], (d) MotionCLIP [51]), as well as robotic skill and affordance learning from human demonstrations ((e) SeeDo [76], (f) CoPa [65], (g) DexFG-Net [77], (h) OKAMI [78]).

Table 1
Robots affordance and value learning.

	Method	VFM	LLM	VLM	Benchmark/Data	Tasks
Skill affordance	CoPa [65]	Owl-ViT, SAM		GPT-4V	VoxPoser	Everyday manipulation tasks
	CLIPort [79]	Transporter		CLIP	Ravens	Language-conditioned tasks
	SayCan [80]		540B PaLM		Everyday Robots	Long-horizon Lasks
	Voltron [81]	ViT		DistilBERT	Franka Kitchen	5 applications within robotics
Keypoint affordance	MOKA [67]	GroundedSAM		GPT-4V	Octo, VoxPoser	Table-top manipulation task with unseen objects
	ReKep [68]	DINOv2,SAM		GPT-4o	VoxPoser	In-the-Wild and Bimanual Manipulation
	KALIE [82]			CogVLM, GPT-4V	MOKA, VoxPoser	Diverse unseen objects and arrangements
Spatial affordance	VoxPoser [15]	OWL-ViT, SAM		GPT-4	RLBench	Everyday manipulation tasks
	RAM [69]	DINOv2 or CLIP	Text-embedding-3	GPT-4V	DROID	Manipulation tasks on common household objects
	RoboPoint [83]	CLIP, ViT-L/14	Vicuna-v1.5-13B		WHERE2PLACE	Language-conditioned manipulation tasks
Human affordance	HRP [84]	DINO, CLIP			Ego4D	Unique manipulation tasks
	HULC++ [85]			GPT-3, MiniLM-L3-v2	CALVIN	Long-horizon manipulation tas

provide robust support for cognition. For example, TCQA [101] enhances contextual awareness in gear assembly, MPKG-WT [102] enables question-answering for wind turbine assembly, and Text2Pickup [86] clarifies ambiguous instructions through question generation to specify task intentions. More details are in Table 2.

Subtask Decomposition. The generative capabilities of MLLMs enable robots to decompose subtasks for Task and Motion Planning (TAMP). Text2Motion [90] scores skills and validates the feasibility of task and motion plans, SayCan [80] integrates VLMs and LLMs to generate plans. Inner Monologue [91] enables closed-loop feedback through internal reasoning with LLMs, while PaLM-E [23] achieves embodied cognition by jointly learning from VQA and robotic data. However, only task decomposition is insufficient for robot cognition, tasks must align with specific and physical constraints—i.e., "grounding". To address this, Grounded Decoding [89], an evolved version of SayCan, combines LLMs' long-term vision and semantic reasoning with local execution. Pg-vlm [59] introduces the PHYSOBJECTS dataset to enhance physical reasoning.

High-level subtask decomposition commonly rely on primitives to implement low-level strategies. Google Robotics has pioneered iterative innovations in this area [80,89,91,92,94]. Primitives including task primitives [59] and skill primitives [31,60] are used as basic operational units. Examples include the pick-and-place meta-operations in CLIPort [79] and two-pose primitives in Transporter [103], which serve as baselines in many subsequent studies. The combination of these atomic skills enables complex tasks and promotes generalization. Primitives can be implemented via trajectory replay [59], teleoperation, or demonstration [60], or through imitation and reinforcement learning. Details will be elaborated in Section 3.3 .

Code Generation. Furthermore, code-focused LLMs (such as Qwen-Code) enable direct generation of executable Python code from language instructions. Early Socratic Models [94] have demonstrated VLM's capability for open-ended reasoning and zero-shot perception. Code as Policies (CaP) [92] generates robot-centric policy code. Some studies [95,96] extends GPT to robots, rapidly parsing human intent and generating code. Liu et al. [98] use GPT-4 to decompose executable robot sequences to enhance autonomy in HRC. GenSim2 [104] even leverages LLMs to automatically generate rich simulations and expert demonstrations.

In particular, for long-horizon tasks, apart from directly generating executable code through API calls [92,97,100,105], it is also possible to first decompose subtasks and then generate code [76,106]. Additionally, using Planning Domain Definition Language (PDDL) to express high-level plans has shown significant improvements [107]. To improve generated program correctness, Progprompt [93] standardizes

Table 2
High-level step-by-step task planning and executable code generation.

	Method	VFM	LLM/VLM	Benchmark	Robot	Tasks
Subtask plan	PaLM-E [23]		PaLM	Language-Table	Everyday Robot	Visually-grounded dialogue
	Pg-vlm [59]	OWL-ViT	GPT-4, PG-InstructBLIP	PHYSOBJECTS	Franka Panda	Table-top manipulation
	ViLA [60]	OWL-ViT	Llama 2-70B, GPT-4V	Ravens	Franka Panda	Long-horizon planning
	SayCan [80]	ViLD	540B PaLM	Everyday Robots	Everyday Robot	Long-horizon tasks
	GD [89]	OWL-ViT	InstructGPT, PaLM	Ravens, CLIPort	Everyday Robot	Rearrangement and mobile manipulation
	Text2Motion [90]		Text-davinci-003 InstructGPT, Codex	TableEnv		TableEnv long horizon manipulation
Code generation	Instruct2Act [64]	SAM CLIP	Text-davinci-003 LLaMA, ChatGPT	VIMABench		Manipulation and reasoning
	Inner Monologue [91]	MDETR	InstructGPT	Ravens, CLIPort	Everyday Robot, UR5e	Rearrangement and mobile manipulation
	CaP [92]	ViLD, MDETR	GPT-3Codex	RoboCodeGen, HumanEval	Everyday Robot, UR5e	Table-top and mobile manipulation
	ProgPrompt [93]	ViLD	GPT-3	Virtual Home	Panda	Household tabletop tasks
	SM [94]	ViLD	GPT-3	CLIPort	UR5	Zero-Shot Multimodal Reasoning
	ChatGPT for Robotics [95]		ChatGPT	PromptCraft, Habitat	Household robot	Zero-shot planning
	RobotGPT [96]		ChatGPT	Pybullet	Franka Panda	Open-ended tasks
	PromptBook [97]	OWL-ViT	PaLM 2-L, GPT-4 Instruct-PaLM		UR5, Kuka2x	Zero-shot planning
	HSR robot [98]		GPT-4 Turbo		Toyota Robot	Zero-shot Basic Tasks
	ReplanVLM [99]		GPT-4V	CoppeliaSim	JAKA Zu 7	Rearrangement manipulation
	TidyBot [100]	ViLD, CLIP	Text-davinci-003		Kinova Gen3	Household cleanup

Table 3
Robots learning from demonstration.

Model	Structure	Problem	Benchmark/Baseline	Input	Output
SeeDo [76]	SAM2+GPT-4o	Code Generation	CaP	Human demo videos	Executable code
OKAMI [78]	GPT-4V, SLAHMR	Humanoid robots manipulation	ORION, Robosuite	Video of human demonstration	Manipulation policy
R3M [108]	ResNet50+DistilBERT	Universal Visual Representation	Ego4D, MetaWorld	Visual Embedding and Proprioception	Action vector
R+X [109]	DINO+Gemini	Retrieval and Execution	R3M	Current RGB-D Observations	6-DoF action
RT-Trajectory [110]	PaLM-E	Robotic Task Generalization	RT-1	Drawings or videos	Trajectory token
Gen2Act [111]	Gemini+VideoPoet+Tap-vid	Behavioral cloning	Vid2robot	Instruction and observations	Trajectory
Egomimic [112]	ACT-based	End-to-end imitation learning	ACT	Hand pose and proprioception	SE(3) pose prediction

prompts and assertion checks. Some works utilize Chain of Thought (CoT) prompting to guide LLMs reasoning systematically. For example, Instruct2Act [64] adds "Think step by step" in prompts for limited context examples, and ViLaIn [107] employs CoT to enhance Corrective Re-prompting.

3.2.3. Learning from demonstration

Learning from human demonstrations is undoubtedly a direct and efficient method of skill acquisition, as it closely aligns with human cognitive perspectives. Egocentric human videos provide ideal resources for expanding passive datasets. As mentioned above, one approach is learning "affordances" from human videos. Another promising direction is language-driven joint learning with videos, as shown in Table 3.

R3M [108] established a baseline for many subsequent works [81, 109] by pre-training reusable visual representations through temporal contrastive learning and video-language alignment. R+X [109] further leverages VLMs to learn skills from unlabeled videos without training. Conditional signals such as 2D trajectories [110] and 3D hand pose [112] enhance interpretability in imitation learning. RT-Trajectory [110] uses human sketches or videos to guide end-effector motions and gripper interactions, offering a novel perspective on learning from humans. This provides a novel approach and has gained increasing attention. EgoMimic [112] further proposes a full-stack framework including data collection and policy learning.

Recent studies explore generalization through language-conditioned video generation [111], direct task planning and code synthesis from human demonstrations [76], and humanoid robots imitation learning from single video inputs [78]. Despite growing attention, egocentric videos often lack detailed action labels, posing limitations in physical interactions. Related challenges and potential solutions will be further discussed in Section 5.1.2.

3.3. Actuation

The action module acts as the "cerebellum" of embodied intelligence, responsible for receiving instructions and executing specific actions through motion planning. Robot learning traditionally relies on supervision such as expert or reward to acquire skills. However, to enhance task efficiency and generalization, leveraging language as an auxiliary tool has become a key focus. Based on low-level strategies, robot learning can be categorized into Language-Conditioned Imitation Learning, Language-Enhanced Reinforcement Learning, and Language-Guided Diffusion Policies, with Tables 4, 5, and 6 showing the details respectively.

Table 4
Language-conditioned imitation learning baselines. Including abbreviations and symbol definitions for Long-Horizon Multi-task Language-Conditioned Policies (LH-MTLC), Current state s_t, Goal state s_g, Latent goal space z in MCIL, task embedding z in BC-Z, and action outputs for end effector position, Euler angles, and gripper state.

Method	Source	Policy	Formula	Input state	Action execution	Core structure
PlayLMP [113]	Lynch et al	GCBC	$\pi_\theta(a\|s_c, s_g)$	Observations and 8-D proprioceptive state	8-DoF action	Seq2seq CVAE
MCIL [114]		GCBC	$\pi_\theta(a_t\|s_t, z)$	Observations, proprioceptive state, instructions	8-DOF action	TransferLangLfP
BC-Z [115]		End-to-end BC	$\pi(a\|s, z)$	Images and task embedding	7-DoF action	ResNet18 +FiLM+FC
Language Table [116]		LCBC	$\pi(a\|s, l)$	Language and Instructions	2D Cartesian point	LAVA
CALVIN [117]	Mees et al	LH-MTLC	$\pi_\theta(a_t\|x_t, x_g)$	Observations, proprioceptive state, instructions, tactile	Cartesian (3) or joint space (7)	Seq2seq CVAE
HULC [118]		LCBC	$\pi_\theta(a_t\|s_t, l)$	Static and gripper camera frame, Instructions	7-DoF action	Seq2seq CVAE
HULC++ [85]		LCBC	$\pi_\theta(a_t\|s_t, l)$	Static and gripper camera frame, Instructions	7-DoF action	HULC+VAPO

3.3.1. Language-conditioned imitation learning

Imitation learning (IL) trains models to derive strategies from demonstrations [113] by collecting large volumes of task data. Behavior cloning (BC), as a form of IL, employs supervised learning to predict actions $f(s) \to a$ from behavioral data $D = \{s, a\}$. IBC [119], BeT [120], and ACT [121] are key baselines in imitation learning.

From IL to LCIL A key aspect of imitation learning is scalable task definition. PlayLMP [113] learns control methods in latent space through "play data", autonomously discovering task semantics without task labels. Lynch et al. [114] extended image goal-driven LfP to LangLfP driven by images or language goals. These works inspired Language-Conditioned Imitation Learning (LCIL), where the flexibility of natural language breaks semantic barriers in human–robot interaction.

LCIL Baselines. With the rise of LLMs, research has increasingly focused on leveraging MLLMs to implement LCIL [114–116,118,122]. These policies, expressed as $\pi_\theta(a_t|s_t, l)$, take the current state $s_t \in \mathcal{S}$ and free-form language instructions $l \in \mathcal{L}$ as conditions to output actions $a_t \in \mathcal{A}$. Within this framework, research has evolved to target specific conditioning types: goal-conditioned behavior cloning (GCBC) $\pi(a|s, g)$ [113,123,124] for image-driven tasks; video-conditioned policies $\pi(a|s, V)$ [76,111,125] for video-driven imitation learning; and latent-conditioned policies $\pi(a|s, z)$ [114,115] for task execution guided by latent variables.

A series of models have emerged under these paradigms. Multicontext imitation learning (MCIL) [114] has facilitated many baselines for LCIL. For example, BC-Z [115] provides early benchmarks focusing on zero-shot or few-shot conditions, and CALVIN [117] combines multimodal high-dimensional inputs for long-horizon and multi-stage tasks. HULC [118] implements hierarchical language-conditioned agents, and its successor HULC++ [85] further integrates task-agnostic control and object-centric semantic understanding. Subsequently, numerous works have emerged building on these baselines. These studies have not only enabled robots to execute novel instructions but also advanced end-to-end visuo-lingo policies, marking a significant step toward scalable systems.

3.3.2. Language-enhanced reinforcement learning

Reinforcement learning (RL) optimizes policies through agent-environment interactions to maximize rewards. However, it faces low sample efficiency and reward design difficulty. LLMs offer new solutions to these challenges. Previous studies [137] have demonstrated similarities and potential connections between RL and language modeling in terms of Markov and sequential nature. LLMs provide innovative approaches for zero-shot reward generation and generalization [128,130,131], reward function optimization [127,129], long-sequence processing [133,138], and sparse reward problems [135,139]. The generation and reasoning capabilities of LLMs to enhance RL can be summarized into four roles: information processor, generator, reward designer, and decision-maker [140].

LLMs as Reward-Designer In traditional RL, manual reward design requires deep domain expertise and task-specific customization. However, the semantic richness of natural language makes it an ideal bridge between modular rewards and goal tasks. Palo et al. [126] use VLMs as internal reward models. L2R [127] employs an LLM-generated reward functions. Rocamonde et al. [128] calculate zero-shot rewards based on the cosine similarity between state and task embeddings. LIV [132] introduces Language-Image Value Learning to generate multimodal dense rewards from videos, reducing the cost of data collection and fine-tuning. To achieve human-level reward design, some works consider self-optimization [129], human feedback [130], and iterative optimization [131].

LLMs as Decision-Maker Besides, pre-trained language models can also assist decision-making. Direct decision-making leverages sequence modeling to translate language instructions into motion skills, and predict actions or trajectories [133–136,138].

Decision Transformer (DT) [133] pioneered a novel approach of using GPT-like models to learn states, actions, and rewards jointly. Building on the DT framework, LaMo [135] integrates LLMs into offline RL with LoRA fine-tuning. For dynamic navigation tasks, RE-MOVE [134] combines LLM's zero-shot generalization with language feedback to adapt to real-time changes. Concept2Robot [136] uses video classifiers as proxy rewards to generate trajectories from large-scale video data. These innovations significantly advance LLM applications in RL, providing more efficient and flexible solutions for robotic operations.

3.3.3. Language-guided diffusion policy

Language-guided diffusion policies lie in Denoising Diffusion Probabilistic Models (DDPM) and Denoising Diffusion Implicit Models (DDIM). Their powerful generative capabilities present a promising solution for end-to-end action modeling.

Chi et al. [141] pioneered the representation of visuomotor policies as conditional denoising diffusion processes to generate robot behaviors. This work has inspired a new wave of policy learning, with many notable studies utilizing diffusion models to directly generate action sequences [142–145], or even predict future states [145,151]. For instance, Ha et al. [146] proposed a "Scaling Up and Distilling Down" framework that combines data generation and policy learning stages to automatically produce robot trajectories based on language instructions. 3DDA [142] combines 3D scene representation with action diffusion to generate 3D trajectories, while PoCo [143] achieves general skills through composite diffusion policies. MDT [144] designs a GPT-diffusion policy to predict action chunks from the current state and latent goals. Octo [147] trains a diffusion head to decode consecutive action chunks.

Table 5
Language-enhanced reinforcement learning.

	Method	Policy	Task/Challenge	MLLM	MLLM for	Environment
Reward designer	Di Palo.et al [126]	BC	Sparse-reward	FLAN-T5,CLIP	Generate a set of sub-goals	MuJoCo
	L2R [127]	MJPC	Bridge the gap	GPT-4	Define reward parameters	MuJoCo
	VLM-RM [128]	DQN, SAC	Zero-shot rewards	CLIP	Compute the reward as cosine similarity	
	Song.et al [129]	PPO	Self-refinement	GPT4	Automated reward Function designer	Isaac Sim
	Eureka [130]	PPO	Zero-shot reward	GPT-4	Generate human-level rewards	Isaac Gym
	Text2Reward [131]	PPO or SAC	Zero/few-shot dense reward	GPT-4	Generate rewards and allow human feedback	MANISKILL2, MetaWorld
	LIV [132]	BC	Goal-conditioned reward	CLIP	Encode multimodal value functions.	MetaWorld, FrankaK-itchen
Decision maker	DT [133]	DT	Short-sighted behavior	GPT	Autoregressively model token in a sequence	Atari, Gym
	RE-MOVE [134]		Zero-shot navigation	GPT-3.5, Llama-2	Convert instructions into action sequence	
	LaMo [135]	DT	Limited data,sparse reward	GPT-2	Predict the future action	MuJoCo, Atari
	Concept2Robot [136]	DDPG	Learn from Human	BERT	Encode instruction and image features	

Table 6
Language-guided diffusion policy.

Model	Core structure	Problem	Input	Output	Robot
Diffusion Policy [141]	DDPM	Diffusion policy	Observation and proprioception	Action Sequence	UR5, Panda
3DDA [142]	CLIP +3D Diffuser Actor	Conditional diffusion probabilistic model	Instruction and 3D scene	Trajectory	Franka
PoCo [143]	Diffusion policy Composition	Heterogeneous Robot Learning	RGB, tactile, point cloud, language	Trajectory	Franka
MDT [144]	CLIP+Voltron+ Perceiver	Core-based Diffusion Policy	Observations and multimodal goals	Action chunk	Franka
ARDuP [145]	SAM+Diffusion Model.	Text-conditioned video generation	Instruction and latent representation	Action sequence	
Scaling up [146]	CLIP+Diffusion policy	MTLC visuomotor policy	Visuo-lingo-motor data	Action Sequence	UR5e
Octo [147]	T5-base	Diffusion policy	Observation and instructions	Consecutive action chunk	9 robots
ALOHA Unleashed [148]	DDIM	Diffusion policy	4 Images and proprioception	Next 50 actions chunk	ALOHA robot
RDT-1B [149]	SigLIP+T5-XXL	Robotics Diffusion Transformer	Visuo-lingo-motor data	Denoised action chunk	ALOHA robot
$\pi 0$ [150]	PaliGemma VLM	Diffusion policy	Visuo-lingo-motor data	Consecutive action chunk	7robots

In addition, diffusion models also show potential in bimanual manipulation. Examples include ALOHA Unleashed [148], which predicts future action chunks via diffusion policies, $\pi 0$ [150], which employs a diffusion-style flow matching loss to generate continuous action distributions, and RDT-1B [149], which represents the largest diffusion-based foundation model for robotic manipulation, it has set new benchmarks in dexterity, zero-shot and few-shot generalizability. So far, diffusion strategies continue to evolve rapidly, driving progress in robotic autonomy.

4. Toward spatial intelligence-driven revolution

Spatial intelligence enables robots to understand within 3D space, this endows robots with human-like "intuition", enhancing more effective interaction and collaboration. This section, as a response to ***RQ2***, outlines pathways toward human-like intelligence and platforms for achieving spatial intelligence.

4.1. Pathways to human-like intelligence

To acquire more robust spatial skills, robot learning has undergone a gradual evolution. Traditional methods such as imitation learning and reinforcement learning are facing challenges of high costs and intensive labor. This has driven the exploration for more efficient solutions: *Could a robust multi-task approach be trained to achieve zero/few-shot abilities across new tasks?* It is within this context that embodied intelligence within MLLMs as the core technology emerged. Building on general capabilities, Sections 4.1.1–4.1.5 progressively demonstrate approaches from direct utilization and instruction fine-tuning to the enhancement of domain-specific performance. The pathway of this paradigm is illustrated in Fig. 6 and Table 7.

4.1.1. Out-of-box usage

With excellent understanding and reasoning capabilities, MLLMs can be applied "out of the box" to various robotic tasks without requiring task-specific data or additional training. This direct application represents a straightforward way to integrate MLLMs into robots.

Recent innovative studies have demonstrated the practical feasibility of this approach. Manipulate-Anything [161] aims to enable robots to handle arbitrary static objects in real-world scenarios. By leveraging GPT-4V and Qwen-VL for task decomposition and multi-view selection, it generates successful trajectories containing language instructions, RGB-D frames, and 6-DoF grasp poses. Meanwhile, Open6DOR [162] utilizes GPT-4V's 3D perception and simulation capabilities to generate target object grasp poses and establishes an open benchmark for 6-DoF object rearrangement. These developments showcase the extensive potential of MLLMs in robots.

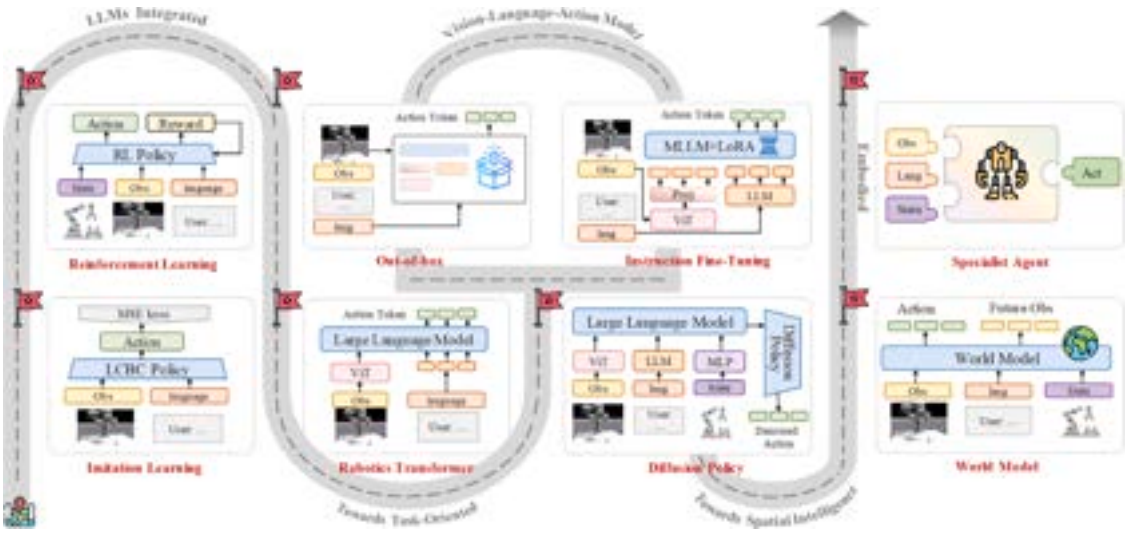

Fig. 6. Pathways to more efficient skill learning and human-like intelligence.

4.1.2. Pretrain and instruction tuning

Although out-of-the-box Usage offers convenience, their performance often falls short due to a lack of task-specific optimization. To address this, fine-tuning becomes essential. However, given the large parameter sizes of MLLMs, a practical strategy is to leverage pretraining on diverse datasets. This pretraining generally follows two main directions. The first involves **domain-agnostic pretraining**, which includes Vision-Language Pre-training [32,158]; Masked Visual Pre-training (for visual representation [169,170] and for decision making [171]); Generative Pre-training (video prediction pre-training [111,122,164]; and Image-Editing Pre-training [123]). The other focuses on **goal-oriented pretraining** specific to downstream robotic tasks, including Sensorimotor Pre-training [172]; 3D Occupancy Task Pre-training [159]; and Robotic Visual Representation Pre-training [173], where MCR [174] enhanced and visualized manipulation-centric visual representations by retraining the R3M [108] on the large-scale DROID dataset.

Notable examples like RT-2 [32], OpenVLA [155] and RoboFlamingo [158] have demonstrated that co-finetune is more efficient than retraining from scratch. These models were initially trained on general visual-language datasets such as LLaVA 1.5 and VQAv2, followed by further fine-tuning on robot-specific datasets like OXE [175] and BridgeData V2 [176]. Despite using predominantly robot-agnostic pretraining data, they exhibited remarkable zero-shot and cross-modal transfer potential.

Especially, unlike standard fine-tuning, language supervised fine-tuning is termed *Instruction Tuning* [19,35] or *Visual Instruction Tuning* [157]. Parameter-efficient fine-tuning techniques, such as LoRA [177] and LCBC+LoRA [178], are widely adopted [155,156,163] in robot control, reducing extensive new demonstration data while improving training efficiency.

4.1.3. Robotics transformer and VLA

VLAs typically take RGB-D observations and language instructions as inputs, generating diverse action representations as outputs. For quadruped robots, actions are represented as joint angles (e.g., Prompt2Walk). For robotic arms, the most common form is 7-DoF actions (`x, y, z, roll, pitch, yaw, gripper`) [30,61,155,158], representing 6-DoF end-effector pose and gripper state, which are often discretized into 256 action tokens (e.g., `(1,128,91,241,5,101,127)`) [32]. Other representations include end-effector waypoints or trajectories [68,161,162], executable text commands specifying actions and positions (e.g., "`rotate <r>` a_r `</r> degrees`", where rotational action is denoted as a_r) [157], and continuous-time sequences of action chunk [147–150]. These diverse output modalities enable VLAs to adapt to various robotic tasks effectively.

Google's Robotics Transformer series [30,32,152,153,175] represents a quintessential example of VLAs. MOO [61] extends the capability to open-world manipulation, enabling generalization to new semantic concepts. RT-2 [32] further utilizes PaLI-X and PaLM-E to enhance policy generalization. To develop generalist "X" robot policies adaptable to diverse robots and tasks, 21 institutions across the globe collaborated to collect the large-scale OXE dataset [175], and re-train high-capacity RT-X. As task semantics grew more complex, RT-H [153] introduces finer-grained phrases for low-level actions and represented language motions as an intermediate layer to bridge the gap between tasks and actions.

Recent studies have refined VLA models further. For example, OpenVLA [155], based on the LLaVA architecture, optimizes dual visual encoders and improves policy precision by discretizing actions into language model tokens. BiVLA [154] introduces an out-of-the-box system for bimanual operations, while TinyVLA [156] addresses inference speed and data efficiency issues. Most of these models rely on Transformer-based architectures. Beyond these advancements, recognizing the strong performance and efficiency of state-space models like Mamba in sequence modeling, Liu et al. [160] introduced Robomamba, this provides a novel perspective for advancing VLA models.

4.1.4. World models for robots

World models serve as the core for achieving spatial intelligence. Spatial intelligence emphasizes the ability to reason and operate in 3D spaces, while the world model provides an intrinsic environmental representation. By integrating multimodal data, world models play a crucial role in fields such as mixed reality and autonomous driving. World models aim to approach human-like intelligence by simulating future states, and are considered as the foundation of physical reasoning [179]. In recent years, the world model for robotics has garnered significant attention. Emerging research such as ByteDance's GR-1 [122] and GR-2 [164], demonstrates end-to-end video-language-action models using GPT-style frameworks to generate actions and predict future video sequences simultaneously. Meanwhile, diffusion models have become a key component with their powerful generative abilities, particularly in text-conditioned video generation. For instance, 3D-VLA [163] integrates diffusion models on top of 3D-LLM to enable multimodal generation including RGB-D and point cloud generation. However, this remains a challenging task due to dynamic scenes. Some studies have explored areas such as enhancing interpretability by Neuro-Symbolic Predicates [180] and establishing unified task

Table 7
The key technology of embodied intelligence, including robotic transformers, Vision-language-action model, Out-of-box usage, and world model.

	Method	VFM/LLM/VLM	Policy	Baseline	Input	Output
RT	RT-1 [30]	FiLM EfficientNet	RT-1	RT-1	Observation and instructions	11D actions
	RT-2 [32]	ViT, PaLI-E, PaLI-X	RT-1	RT-1	Observation and instructions	7D action tokens
	MOO [61]	Owl-ViT, FiLM EfficientNet	LCBC	RT-1	Images and language instructions	7D action tokens
	Q-Transformer [152]	FiLM EfficientNet	Offline RL	Manual dataset	Observation and instructions	Q-value of action
	RT-H [153]	ViT, PaLI-X	LCIL	Kitchen dataset	Image, task, and action query	Action token
VLA	Bi-VLA [154]	Qwen-VL	No train		Observation and user request	Executable code
	OpenVLA [155]	SigLIP, DinoV2, Prismatic-7B	LoRA Finetuned	OXE, BridgeData V2	Observation and instructions	7D action tokens
	TinyVLA [156]	Pythia	LoRA Finetuned	MetaWorld	Observation and instructions	6D action
	LLaRA [157]	GPT-4, LLaVA-1.5-7B	LCBC+Instruction Tuning	VIMA, inBC, D- inBC	Observation, task, and previous actions	Textual actions
	RoboPoint [83]	CLIP, Vicuna-v1.5	Instruction Tuning	WHERE2PLACE	Observation and instructions	3D action points
	Roboflamingo [158]	LlaMA, GPT-4, OpenFlamingo	GCBC	CALVIN	Task and 2 camera views	7D action tokens
	RoboUniView [159]	ViT, UVFormer	GCBC	CALVIN	Task and multi-camera views	7D action tokens
	RoboMamba [160]	CLIP, Mamba	Instruction Tuning	LLaVA 1.5, RoboVQA	Image and language question	6-DoF EEF poses
Out-of-box	CoPa [65]	Owl-ViT, SAM, GPT-4V	No train	Real-world data	Observation and instructions	6-DoF EEF poses
	VoxPoser [15]	Owl-ViT, SAM, GPT-4	No train	RLBench	Observation and instructions	Sequence of 6-DoF waypoints
	ReKep [68]	DINOv2, SAM, GPT-4o	No train	VoxPoser	Observation and instructions	Sequence of 6-DoF poses
	MA [161]	GPT-4V, Qwen-VL	No train	RLBench	Task goal and multi-views images	6-DoF EEF poses
	Open6DOR [162]	GroundedSAM, GPT-4V	No train	Synthetic dataset	Observation and instructions	Robot motion trajectory
World Model	3D-VLA [163]	Flan,T5XL,BLIP2	LoRA Finetuned	OXE, RH20T	Interaction token with 3D sence	Image, pointcloud, action
	GR-1 [122]	ViT, CLIP	LCBC	RT-1, HULC, R3M, CALVIN	Instructions, video frame, state	Images, action trajectories
	GR-2 [164]	VQGAN, cVAE		GR-1, RT-1, HULC, RoboFlamingo	Instructions, video frame, state	Images, action trajectories
	RoboDreamer [151]	T5-XXL		UniPi, AVDC, RLBench	Language and multimodal Instructions	Video and actions
	EVA [165]	CLIP, Vicuna-v1.5, ChatUniVi	Instruction Tuning	EVA-Bench	Observation and instructions	Videos, text responses
	PIVOT-R [166]	CLIP, LLAVA	Imitation Learning	BC-Z, Gato, RT-1, Octo, GR-1	Instructions, observation, state	Waypoint image, EEF action
	DINO-WM [167]	DINOv2		Dreamerv3, AVDC	Current and goal observation	Action sequence
	WHALE [168]	ST-transformer	Fine-tuning	OXE, Meta-World	Observation and action subsequences	Observation predictions

benchmarks in EVA [165]. The recent work WHALE [168] improves the world model in OOD regions through behavior-conditioning and retracing-rollout, advancing the application of world models in physical operations. This highlights the potential of spatial intelligence to unlock new possibilities for physical interactions.

4.1.5. Embodied specialist agent

Generalist Large Models for Daily Tasks. Cross-task, cross-scenario, and cross-entity generalization are long-term goals for achieving AGI. Pioneering models like Gato [181] represent multimodal, multi-task generalist policies that span diverse applications from Atari games to complex robotic operations. Building upon such frameworks, RoboCat [182] demonstrates self-improving capabilities by leveraging heterogeneous robot data to master new skills iteratively.

Beyond task generalization, cross-entity generalization is also critical. It focuses on adapting to the heterogeneity of robot data since input and output dimensions may vary based on embodiment-specific implementations. Breakthrough approaches like Octo [147], support flexible task definitions and quickly adapt to new robot setups. $\pi 0$ [150] collects a diverse cross-entity dataset with seven different configurations, achieving a novel generalist robot policy using the PaliGemma model and flow matching to enable control within different action spaces. HPT [183] further incorporates proprioception and vision, including real robots, human videos, and simulation data. By exploring scaling behaviors across heterogeneous domains, these approaches are progressively dismantling barriers to versatile robotic intelligence.

Specialist Small Models for HRC Needs. Undoubtedly, the above works have significantly pushed the boundaries of generalist agents, and extended basic primitives like pick-and-place to more complex and challenging skills such as pouring and cooking. However, most of these efforts only involve daily life tasks with limited application scope.

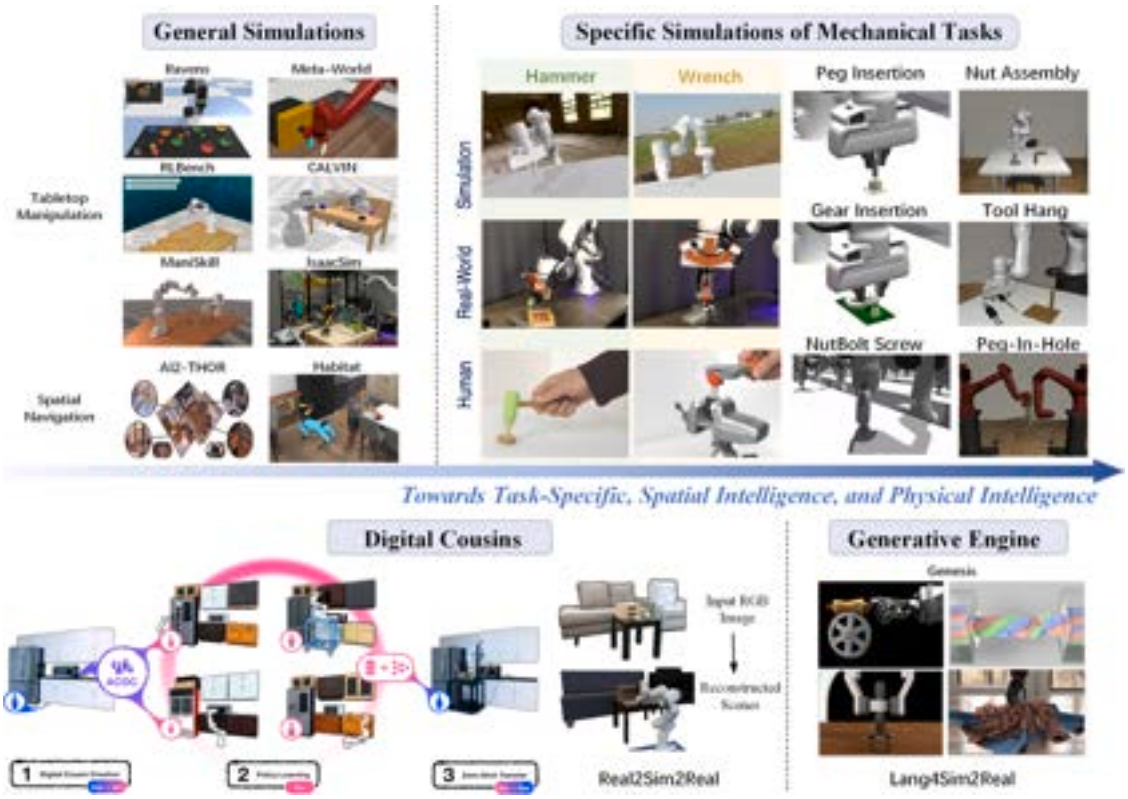

Fig. 7. Platforms for low-cost and multi-skill learning, including general manipulation and navigation simulators, task-oriented specific environments, Genesis for physical intelligence, and digital cousins for spatial intelligence.

There remains a significant research gap for more diverse tasks, particularly in production and manufacturing. A few studies have started to address this domain, such as industrial-like scenarios involving mechanical operation. Examples include tools such as *Hammer* and *Wrench* in PoCo [143], actions such as *Gear Insertion* and *Screw* in Isaac [184], and tasks such as *Nut Assembly* and *Peg-In-Hole* in Robosuite [185] as shown in Fig. 7. Although these explorations are in their early stages, they represent a promising beginning.

Thus, it is emphasized that specialist or "small models" designed for targeted tasks are also important alongside the generalist large models for daily tasks. Particularly, developing tailored models on the strengths of large models represents a promising direction. It is hoped that this work will draw attention to this perspective.

4.2. Platforms and environments

4.2.1. Datasets, simulators and benchmarks

With the rise of embodied intelligence, numerous resources and platforms have emerged for training specific models.

- **Datasets**: Currently, robotic learning remains data-driven, with scaling laws [186] revealing a roughly power-law relationship between policy generalization and environment-object pair diversity. Recent advancements focus on training robot policies through diverse datasets, emphasizing large-scale, multimodal, and cross-entity approaches, integrating various inputs like RGB, depth, point clouds, trajectories, and linguistic annotations [30,175,176,187], even force, tactile, and auditory data [188]. This trend spans from desktop manipulations to complex kitchen and outdoor environments, supporting transfer learning across varied robots—from traditional robotic arms to dexterous hands, legged robots, and musculoskeletal systems [189], enabling cross-robot transfer learning.
- **Simulators**: Core physics engines (e.g., ODE, Bullet, PhysX, MuJoCo) and 3D engines (e.g., Unity) provide support for advanced simulators. Early general simulators like Gazebo, V-Rep, Webots, and Pybullet facilitated basic robot modeling through rigid dynamics. In realistic simulations, two primary focuses are manipulation and navigation. Where manipulation simulators mainly emphasize desktop tasks, such as SAPIEN-powered ManiSkill families [190,191], the GPU-clustered ecosystem (from IsaacGym [184] to IsaacSim and IsaaLab [192]), and is evolving toward general daily tasks and humanoid robots [185,193]. Navigation simulators usually target indoor environments [194–197], some of them integrate VR interfaces [198,199] to enable remote operations. Habitat 3.0 [200] supports human simulation and enables humans-as-agents to interact with robots and environments, paving the way for embodied human–robot interaction.
- **Benchmarks**: These simulators not only provide diverse environments but also integrate classic algorithms and benchmarks for robot learning, ranging from offline imitation learning [201], vision-guided behavior cloning [103], state driven reinforcement learning [202,203] to language-conditioned tasks [116] and multimodal long-horizon tasks [117]. These lay a foundation for training and evaluation.

4.2.2. Digital cousin in cyber space

Simulators along with their datasets and benchmarks provide platforms for agents to rapidly learn multiple skills. This transfer from physical to virtual world highlights the significance of learning in cyberspace and has driven the exploration of spatial intelligence. However, most platforms are limited to tabletop tasks. Although some simulators have presented industrial-like environments and mechanical scenarios, for specialized applications like manufacturing, creating corresponding environments is undoubtedly time-consuming and labor-intensive. Extending to these complex tasks remains a key challenge.

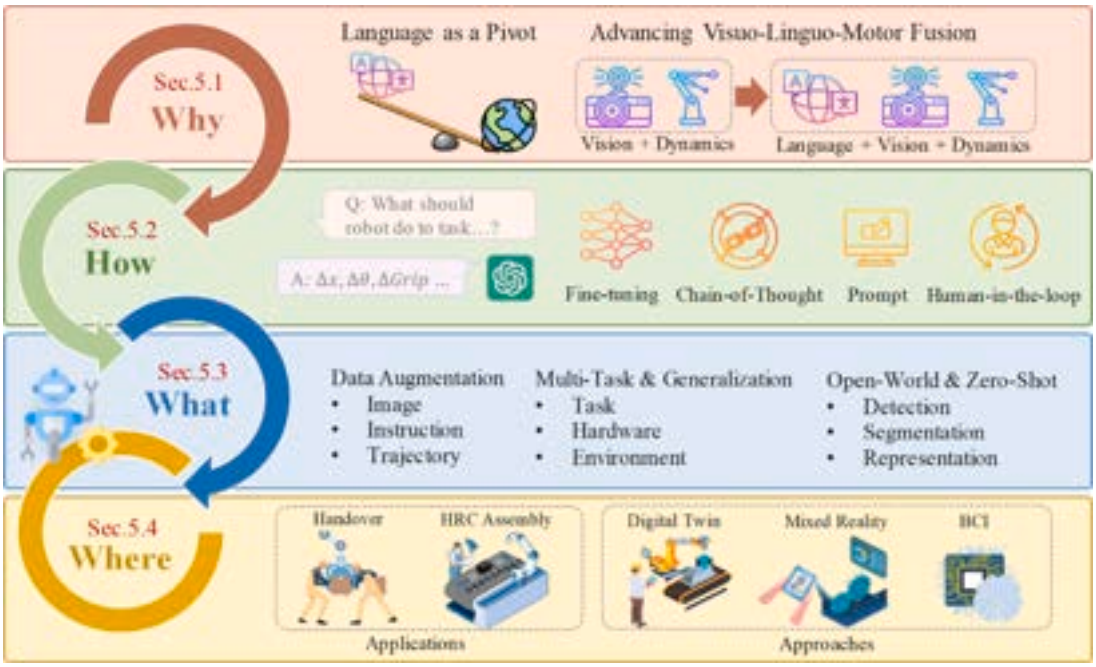

Fig. 8. Overview of the critical discussions of MLLM-enabled HRC.

To bridge the gap between simulation and practical applications, researchers have explored learning in cyberspace. One promising approach is **Real2Sim2Real**, where methods like domain randomization and system identification reduce sim-to-real transfer bias. However, studies on Real-Sim are relatively limited and primarily rely on scene scanning and reconstruction, both require extensive data collection.

The Institute for Human-Centered AI at Stanford University recently introduced the concept of "*Digital Cousins*" [8] and the ACDC (Automated Construction of Digital Cousins) method. Unlike the well-known "*Digital Twins*", it emphasizes spatial computation capability rather than high-fidelity visualization. This approach has demonstrated the ability to construct interactive virtual environments from a single RGB image and skill learning within these environments. This paves the way for low-cost, rapid learning and zero-shot deployment of robots.

Additionally, **Lang4Sim2Real** also presents a potential solution. It not only uses natural language as a bridge to narrow the sim-to-real gap by employing unified textual descriptions, but also leverages the generative capabilities of LLMs to assist in scene reconstruction. A typical example is the latest general physics engine, Genesis [204], which supports language-based simulation generation, offering a bright future for constructing task-specific environments. Prospectively, these methods show great promise for rapidly building cyberspaces for specific applications, offering unprecedented opportunities for the challenges of high-cost training and spatial-skill learning.

5. Discussions

This section explores key topics essential for multimodal HRC from four perspectives, including language integration, task optimization, challenges and practical applications, as illustrated in Fig. 8.

5.1. Why language-multimodal is essential

5.1.1. Harnessing language as a pivot

Enhanced Natural Interaction. Language is a pivotal enabler for expressing high-level intentions, actions, and commonsense knowledge about the world. The language modality offers an intuitive to helps break semantic barriers. This natural interaction not only streamlines collaboration but also enhances safety by enabling robots to better anticipate human intentions and respond appropriately. Furthermore, it plays a crucial role in reducing situational ambiguity [86,205].

Advanced Autonomy and Intelligence. Secondly, language serves as semantic supervision, providing task descriptions and prior knowledge that assist agents in exploring policy and improving training. Language-driven strategies have been proven effective in enhancing robots to perform long-horizon tasks [80,117,158], optimize control strategies [127,128,130,131], and even generate executable codes [92,94,95] and actions [32,147,155]. These advancements contribute to more adaptive systems that address complex real-world applications.

Broader Generalization and Adaptability. In addition, language supervision has greatly promoted the development of open vocabulary models. This generalization ability is crucial for adapting to new challenges without the need for extensive retraining. Some out-of-box MLLMs for perception and planning have demonstrated excellent performance in zero-shot and few-shot generalization, as evidenced in [67,161], this enables robots to quickly adapt to diverse and unpredictable real-world applications.

5.1.2. Advancing visuo-lingo-motor fusion

The integration of multimodal inputs has catalyzed a clear hierarchical evolution in robot learning. As shown in Fig. 9, this trend signifies an advancement toward the *"visuo-lingo-motor"* paradigm of embodied intelligence. Specifically, *"visuo"* and o represent visual observations, *"lingo"* and l denote language instructions, and *"motor"* and s refer to motion or state (e.g., joint and end-effector proprioception). Robot datasets have developed from single observation-action trajectories $\tau_i = [(o_0, a_0), (o_1, a_1) \dots]$ to multimodal representations incorporating linguistic context $D(l, a) = (\tau_1, l_1), (\tau_2, l_2), \dots (\tau_N, l_N)$. Algorithmically, it has also evolved to more comprehensive methods, which can be summarized into four key paradigms:

- Visuo/Motor: (1) "vision → action" $\pi(a|o)$, robots output actions based on visual observation, as in [141]. (2) "state → action" : $\pi(a|s)$, For example, DT [133] use states to predict actions at the next time.
- Visuo-Motor: "vision + state → action": observations and proprioceptions (e.g., joint positions) are integrated to output actions [121,182,183,201].

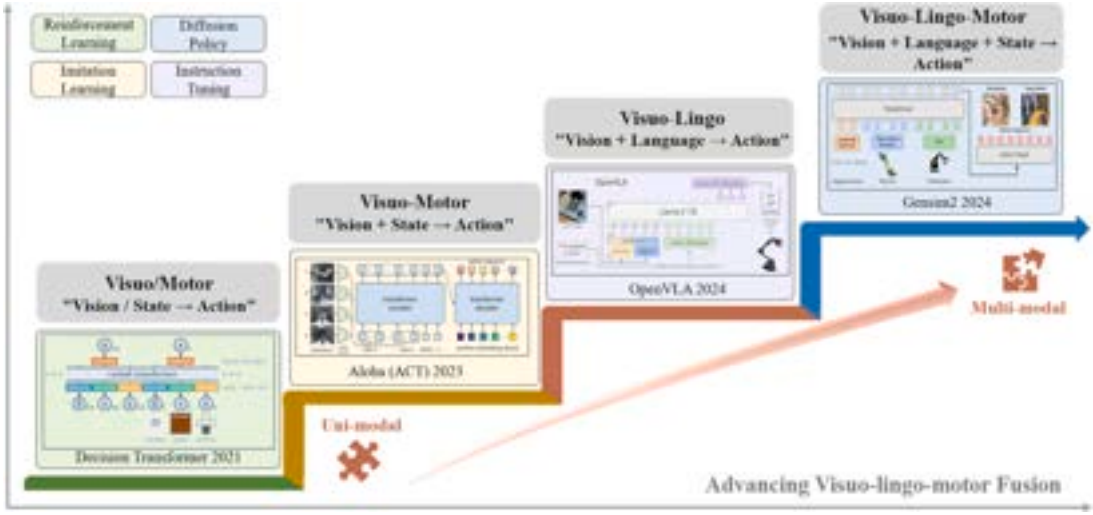

Fig. 9. Advancing "visuo-lingo-motor" fusion: evolution from uni-modal to multimodal that includes language as a unified bridge and physical state as proprioception. *Source:* Adapted figures from [104,121,133,155].

- Visuo-Lingo: "vision + language → action" $\pi(a|o,l)$, also known as language-conditioned visuomotor policy [146], this paradigm predicts actions from observations and instructions. Most VLAs follow this structure, such as [32,155,158]
- Visuo-Lingo-Motor: "vision + state + language → action" $\pi(a|s,o,l)$, where robots holistically integrate visual, linguistic, and physical inputs, such as in [104,122,146,147,150].

The above evolutionary trends highlight the pivotal role of integrating proprioception and language. On one hand, proprioception equips robots with real-time physical state awareness, such as joint positions [121,150], end-effector poses [122,206], and forces and torques [147]. This allows robots not only to "see" but also to "feel". Such capability directly impacts dynamic consistency, particularly in high-precision tasks such as screw tightening, grasping fragile objects, and avoiding human collisions. On the other hand, language plays a crucial role in addressing the challenge of robot heterogeneity in large-scale manufacturing, which manifests in visual (e.g., different camera positions), structural (e.g., varying degrees of freedom), and proprioceptive aspects. Language not only expresses task semantics but also bridges different modalities. Studies like [149,150,183] employ encoders to map observations, instructions, and states into a unified latent space, enabling cross-modal alignment and synergistic processing.

5.2. How to improve task-oriented HRC

5.2.1. Optimized fine-tuning

MLLMs have demonstrated remarkable capabilities across various applications, but in specific domains such as machining tasks, generic models often struggle to adapt to downstream tasks due to the lack of domain-oriented training within manufacturing knowledge. Therefore, efficient fine-tuning optimization is essential [207].

Fine-tuning methods include *Prompt Tuning* and *Instruction Tuning*. Prompt Tuning optimizes input prompts to guide model responses without altering model parameters. Examples like CoOp [208] introduce learnable soft prompts for dynamic scenario adaptation. Conversely, Instruction Tuning directly refines model weights to improve task instruction comprehension and execution.

To reduce costs while maintaining performance, efficient fine-tuning strategies concentrate on two main approaches: computational efficiency (e.g., mixed precision training, flash attention, S^2 attention) and parameter optimization (e.g., full fine-tuning (FFT) and parameter-efficient fine-tuning (PEFT)). PEFT methods include Adapter Tuning, Prefix Tuning, Prompt Tuning, and LoRA [177], where LoRA uses low-rank matrices to approximate weight updates. LLaMAFactory [209] integrates advanced efficient training methodologies, significantly reducing memory requirements for fine-tuning. These approaches provide strong support for the application of MLLMs in HRC, enabling optimized performance even in resource-constrained environments.

5.2.2. Tailored prompt design

In personalized reasoning, prompt design has become a key approach to maximizing the potential of LLMs. Compared to fine-tuning which requires retraining parts of the model, it provides a flexible and efficient method, gradually emerging as the preferred approach for optimization. Well-designed prompts can guide LLMs to better align with human emotional behavior and specific task requirements. Particularly in specialized applications like HRC, integrating domain-specific knowledge (such as assembly processes) into the prompts can significantly enhance task execution.

Many studies have addressed the question of prompts elements [97,210,211], covering Few-shot examples, instructions, task descriptions, and contextual information. Notably, CoT incorporates "Think step by step" in prompts to guide incremental reasoning. This trick has significantly improved performance in complex tasks such as multi-turn dialogue and long-horizon tasks [91], action sequence generation [64], and embodied execution [212].

Meanwhile, the importance of prompt design has been increasingly emphasized in research on robots. For example, PromptBook [97] has developed prompts specifically for robot tasks. Numerous works provide rich prompt examples in their appendices, such as multimodal prompts of 17 tasks in VAMA [31], additional robotic arm prompts in LLaRA [157], action generation prompts in AToM-Bot [48], prompts with pythonic examples in Progprompt [93], and prompts specific to different robots [129].

5.2.3. Human-in-the loop

Unlike traditional robot execution, the HRC loop as shown in Section 3.1 in this work specifically incorporates human activity and intent recognition, which is critical for achieving genuine human–robot collaboration. In industrial scenarios with complex and high safety demands, such as precision operations (e.g., nut assembly, chip placement), dynamic object handling, and bimanual coordination, human supervision ensures safety and reliability. Similarly, in tasks requiring human judgment, such as ethical decision-making, creative problem-solving, or medical procedures, human feedback not only rectifies system errors but also provides insights beyond the capabilities of AI.

Human feedback significantly enhances model performance through various forms, including preference expression, voice commands, SpaceMouse intervention [206,213], and even gaze tracking [214]. Moreover, integrating human supervision into imitation and reinforcement learning is also effective. For instance, real-time human intervention to correct robot actions enriches reinforcement learning buffers or prioritizes intervention data, optimizing the learning direction. Such data accelerates the robot's ability to learn correct behaviors, avoid local optima, and expand its state space. Human-in-the-loop involvement addresses task uncertainties and enhances the robot's flexibility and reliability in completing tasks.

5.3. What can contribute to HRC challenges

5.3.1. Data augmentation

Large-scale, high-quality data serves as the foundation of learning-based robot systems, directly influencing their performance and reliability. Given the prohibitive costs of real-world data collection, augmentation methods like simulation synthesis and domain randomization have gained widespread adoption. In recent years, the generative capabilities of MLLMs have provided new approaches to address data scarcity, primarily through the following ways.

Image augmentation is critical for training generalist agents but must avoid altering the relationship of original information and actions. Given this, CACTI [215] leverages Stable Diffusion for image inpainting and scene attribute modification, GenAug [216] generates semantically equivalent observation-action pairs, and ROSIE [58] uses Image Editor with text prompts to insert new objects to enrich visual diversity.

Instruction augmentation aims to improve robot adaptability to task semantics and unseen environments by generating varied instructions. Examples include utilizing GPT-3 to generate augmented detection prompts [58], fine-tuning VLMs on small datasets to create new instruction labels [217], and expanding synonymous instructions or reframing queries using GPT-4 [157,158,178].

Trajectory augmentation focuses on addressing the high cost of data collection. Typically, Ha et al. [146] employ LLM-driven high-level planning to create multi-task language-labeled strategies. RoboGen [218] combines GPT-4 with Genesis simulations to generate diverse task prompts and skills, and GenSim2 [104] uses LLMs to automatically generate simulated environments and expert demonstrations. These methods provide robust support for efficient learning paradigms.

5.3.2. Open-world and zero-shot

The real world represents a dynamic and multimodal environment. In proactive collaborations, agents are required to possess autonomous cognition to handle this complexity. Open-world capability is particularly critical, as it enables robots to rapidly adapt to dynamic tasks, support low-code programming [219], and facilitate quick demonstrations to meet the demands of small-batch, multi-variety assembly.

Most current approaches mainly rely on expert demonstrations and scripted strategies. Yet the development of MLLMs and their inherent emergent capabilities has endowed agents with zero-shot and few-shot learning potential. Advanced open detection and generalized segmentation models in Section 3.1.2 have significantly enhanced environment understanding. Examples like RAM use CLIP and DINO to address data scarcity through out-of-domain learning, and ViLaIn's innovative fusion of visual-linguistic modalities demonstrates exceptional performance in open-set detection and zero-shot planning. These advancements substantially improve the efficiency and adaptability in unseen worlds and lay a solid foundation for the future of human–robot collaboration.

5.3.3. Long-horizon task

Long-horizon tasks, characterized by multi-steps and extended temporal complexity, have traditionally been addressed through symbolic and geometric reasoning. However, the advent of LLMs has revolutionized these strategies. Pioneering research has demonstrated LLMs' capabilities in selecting skills [80,90,91], primitives decomposition [31,60]. It also enables novel language-conditioned policies or action heads to predict long-term action sequences.

Recent benchmarks have further advanced long-horizon research. Recognizing the limitations of multi-context imitation learning for complex operations, Mees et al. [117] proposed CALVIN which reframed long temporal state–action streams as goal-conditioned demonstrations. Several works [122,123,158,159] have progressively pushed new boundaries and achieved state-of-the-art performance on this benchmark. HULC++ [85] in combination with LLMs enables multi-tier and long-horizon task execution from abstract language instructions. Furthermore, text-to-video models have also been proven to optimize long-term planning [220], signaling a future where semantic supervision and multimodal large language models could potentially reshape robot execution across dynamic and intricate operational landscapes.

5.4. Where to meet urgent applications

The integration of HRC and MLLMs addresses urgent applications across three levels, as shown in Fig. 10. The following subsections explore in detail and provide examples.

5.4.1. Coexistence: Human-robot handover

In the context of HRC, object handovers have become increasingly critical. Compared to predefined tasks, dynamic handovers present a significant challenge that demands urgent attention. This collaboration can be categorized into two paradigms: robot-to-human (R2H) and human-to-robot (H2R).

In the R2H paradigm, robots are tasked with grasping and transferring objects to humans. Research focuses include ensuring ergonomic comfort [225], and optimizing grasp strategies by observing human-to-human handover patterns [226]. Meng et al. [227] utilize vision-based detection to enable fast and comfortable handovers. However, the high cost of data collection promotes the emergence of affordance-based methods. Examples include vision-based value calculation for functional object features [228] and 3D affordance map prediction for human contact point optimization [222].

Conversely, in the H2R paradigm, prior studies have explored predicting handover points based on human behavior patterns to improve comfort [229–231]. To address the limitation of specific object categories, Yang et al. investigate human grasp classification [232] and visual segmentation [233], achieving reactive H2R handovers for arbitrary objects. Recent efforts have addressed the challenges of benchmarks and realistic physical modeling. HandoverSim [234] introduces a new simulation framework to model human grasping and object release actions. SynH2R [235] generates static, handover-friendly grasp poses and leverages synthetic data for training. Building on these works, Wang et al. [236] integrate 4D point clouds and motion prediction to develop a large-scale environment, GenH2R-Sim, significantly enhancing handover performance.

5.4.2. Collaboration: Smart manufacturing

LLMs in lifecycle of Manufacturing. LLMs have demonstrated significant potential across stages of the product lifecycle in manufacturing, including design, production, and service. In the design phase, LLMs enhance decision-making by integrating domain expertise, with their interactivity reinforcing a human-centered design approach. Liane et al. [237] summarize how LLMs can accelerate computational design and manufacturing (CDaM). In the production and manufacturing phase, LLMs exhibit advanced capabilities in understanding human intent, allowing robots to quickly adapt to new dynamic assembly requirements, especially in non-standard situations or special requests. Relevant applications have been demonstrated such as language commands disambiguation [105], human–robot collaborative navigation [238], unstructured tasks completion [106]. In the management and service phase, LLM-based VQA systems have optimized quality

Fig. 10. Three levels of application. **Coexistence**: in (a) dexterous handovers [221], and (b) contact handover [222]; **Collaboration**: in customized manufacturing, including (c) HRC assembly [223], (d) Virtual Reality [224], and (e) Digital Twins [106]; **Coevolution**: human–robot symbiosis in (f) brain–computer interface and large brain models.

management and reliability in aerospace products [239], and fine-tuned LLMs integrated with domain-specific knowledge have enhanced the ability to respond to engineering queries effectively [207].

MLLMs in Digital Twins and Virtual/Mixed Reality. Integrating advanced MLLMs with Digital Twin (DT) and Virtual/Mixed Reality (VR/MR) offers new opportunities for intelligent manufacturing. This innovation enhances the immersion and interactivity of virtual environments, connecting the physical and virtual worlds to optimize tasks and improve safety. For instance, ProcessGPT [240] showcases the fusion of generative AI and DT. Chen et al. realize autonomous HRC assembly using LLMs and DT, Sun et al. [241] achieve LLM-empowered Digital Twins for intelligent agents. Additionally, LLM-based language-motion models [50,52] facilitate 3D human modeling and human digital twins, advancing HRC in practical applications.

What is more, the Industrial Metaverse [242] is revolutionizing the visualization and interaction between humans and the physical world, with hardware like HTC VIVE, HoloLens, and Apple Vision Pro, and platforms like Nvidia Omniverse and CoppeliaSim. These developments provide robust support for digital transformation, bringing us closer to human-centric smart manufacturing.

Embodied Intelligence in Manufacturing. Embodied intelligence in manufacturing integrates physical robots with advanced AI to perform complex industrial tasks. Although this field remains a novel and emerging area, it has already demonstrated remarkable progress in robotic assembly, contact-rich manipulations, bin-picking, and embodied planning, showcasing its potential to revolutionize industrial automation. For example, Masaru et al. [243] utilizes VLM to identify mating holes for zero-shot peg insertion. A generalist robot agent, GR-2 [164], excels in generalizing to unseen object in industrial bin-picking including transparent, deformable, and reflective objects, (e.g. metal hammers, etc.). Fan et al. [244] propose an LLM-centric framework for task-process matching and tool path design. These innovations underscore the significant potential of embodied intelligence in smart manufacturing, indicating a future trend where MLLMs will be increasingly integrated, fostering a new era of efficiency, adaptability, and innovation in industrial automation.

5.4.3. Coevolution: Brain-computer interface

The Brain-Computer Interface (BCI) serves as a vital bridge between the human brain and external devices, and has achieved significant advancements in neuroscience and artificial intelligence in recent years. Applications include EEG-based intent recognition and brain-controlled robots. For instance, NOIR [245] utilizes steady-state visually evoked potential (SSVEP) and motor imagery to decode EEG signals for precise robot control. Additionally, studies have explored industrial applications, such as function block-based brainwave assisted assembly [246] and closed-loop systems with augmented reality feedback [247].

Different regions of the brain are associated with distinct functions, producing brainwaves of varying frequency bands in response to external stimuli. High-gamma band signals closely associated with complex cognitive functions have emerged as a key focus for motor decoding. Beyond this, language decoding - converting brainwaves into synthesized language - shows promise in enabling communication for individuals with language impairments. More importantly, it has catalyzed the integration of BCI with large models. Researches like the Large Brain Model LaBraM [248] have explored universal representations of brain signals. As those models that integrate BCIs and MLLMs evolve and mature, they may unlock and pioneer new possibilities for thought-based communication in cyberspace. In addition, direct brain–computer communication through implanted electrodes further suggests an increasingly intimate human–robot integration, moving toward human–robot symbiosis.

6. Future challenges

As collaborative robots become more autonomous and powerful, ensuring human-centered systems requires adherence to key ISO standards, including the safety standard ISO 10218, the technical specification ISO/TS 15066, and the ergonomics standard ISO 9241-11:2018. This chapter explores challenges and future research perspectives, answering ***RQ3***.

6.1. Hallucinations in VLMs

Hallucination refers to the inconsistency between the generated content and facts, which can lead to errors and safety risks. It primarily manifests in three forms: existence hallucinations (generation of non-existent objects), attribute hallucinations (inaccurate descriptions of object properties), and relational hallucinations (misinterpretation of relationships between objects). The causes of hallucination include noisy training data, lack of diversity, statistical bias, and inherent limitations in the models.

To address these issues, various evaluation metrics and benchmarks have been proposed, such as Hallusionbench [249]. Furthermore, mitigation strategies are continually explored, including pre-correction with negative instructions and post-correction methods. Additionally, Prismatic VLMs reduce the likelihood of "imagining" non-existent objects or attributes to enhance accuracy. Although hallucinations remain a long-term challenge, optimized foundation models show significant potential in improving behavioral reliability and safety, fostering greater trust in human–robot interactions.

6.2. Errors and failures in LLMs

The hallucination primarily involves the generation of content that contradicts reality, but a larger portion of the errors stem from failures in the model's understanding and execution. Therefore, automatically detecting and analyzing failures in robots is crucial. Several studies have investigated LLM planning errors [97,161]. However, reliability issues are still evident. To address this, Sun et al. [250] summarize how LLMs detect and handle errors when using tools and created a multimodal error dataset. Liu et al. [251] develop the RoboFail dataset and provide a crucial benchmark with various robot tasks.

Moreover, human assistance and feedback can significantly enhance task execution. Additional approaches include employing closed-loop assertion checks to analyze LLM-generated code errors [93], utilizing hierarchical summaries of robot experiences for fault explanation and correction [251], and implementing self-correction mechanisms [96]. Unlike other methods focused on high-level task planning corrections, SC-MLLM [252] and AIC-MLLM [253] proposed by Peking University focus on end-effector posture correction, they provide failure reasons and re-predicted actions as well as addressing difficult-to-correct low-level SE(3) posture actions. These methods provide strong support for improving the reliability in complex collaborative tasks.

6.3. Egocentric viewpoint

Despite significant advancements of MLLMs in traditional tasks, current research predominantly adopts third-person perspectives, overlooking the unique value of egocentric perception. First-person viewpoint data inherently captures more nuanced human–environment interactions. Early studies explored egocentric human-object interaction (EHOI) datasets in industry-like environments. Pioneering datasets like MECCANO, Assembly101, and EgoISM-HOI include multimodal (RGB+Depth) and multi-view (1st and 3rd) data representations. Subsequent works such as Ego4D and EgoExo4D expanded the scale and diversity through AR devices. Based on these datasets, Mu et al. [212] created the EgoCOT and EgoVQA datasets for embodied planning and reasoning tasks, advancing egocentric research within MLLMs.

The integration of first-person perspectives with MLLMs shows potential for enhancing multimodal perception and improving robot autonomous reasoning for embodied actions. Several studies have demonstrated the feasibility of this approach, such as the evaluation benchmarks for egocentric perception [254], egocentric navigation [255], and general manipulation skill-learning from human video (in Section 3.2.3). However, current research has yet to deeply explore egocentric "thinking" and reasoning. Consequently, advancing this paradigm represents a critical research direction, holding profound implications for the development of embodied intelligence and the promotion of human–robot collaboration.

6.4. Multi-agent collaboration

Self-organizing multi-agent cooperation is crucial for realizing proactive human–robot collaboration systems. While the capabilities of LLMs have been validated in single-agent task planning, their performance in multi-agent environments remains unclear, especially in task decomposition, allocation, and scheduling. Coordinating multiple agents requires effective communication mechanisms to ensure efficient labor division and resource allocation.

To address these challenges, an effective approach is coalition formation [256]. Additionally, another innovative method in the context of MLLMs is utilizing multi-agent dialogue, as seen in RoCo [257]. These systems have successfully achieved collision-free motion and task planning. Furthermore, robot clusters integrating VLMs and LLMs (with 20+ robots) have shown remarkable abilities to collect data in real-world settings, as demonstrated by AutoRT [258]. However, as system scales expand, particularly in dynamic and unstructured environments, ensuring communication efficiency and reliability becomes a critical challenge, which will be essential directions for the development of self-organizing multi-agent cooperation.

6.5. Tactile modalities

The integration of multimodal data is crucial for interactive capabilities. Although language serves as an effective mechanism, it is not universally applicable. To endow robots with human-like "intuition", it is essential to construct world models capable of accurately modeling, understanding, and reasoning about spatial and physical interactions, including additional information such as vision, force, and touch.

For the significant challenges that remain in representation, mapping, alignment, and fusion, several studies have explored corresponding solutions, such as latent space sharing [40], cross-modal transformation [42], and modality-switching instruction tuning [41]. Pioneering frameworks like ImageBind [259] and Meta-Transformer [260] demonstrate progressive efforts in unified representation, supporting 6 to 12 distinct data modalities. The evolutionary trajectory of these methods—whether vertically scaling parameters within a single modality or horizontally expanding universal representations—reflects the advancement of generalized perceptual capabilities.

Moreover, with the growing application of XR (Extended Reality) and digital twin in HRC, the demand for multimodal feedback in remote collaboration is increasing. However, current research mainly focuses on visual experience, while immersive remote and virtual reality requires richer modalities. In real-world applications, particularly in precise industrial tasks, tactile and force feedback are also indispensable. Recent studies have increasingly emphasized the role of tactile and force feedback in robots [43,143,188]. These integrations are expected to catalyze intelligent manufacturing and drive unprecedented technological advancements. Such advancements are expected to drive interactions closer to the real physical world, representing one of the key challenges that urgently need to be addressed.

6.6. Physical intelligence

Despite their strong reasoning capabilities across various domains, current LLMs face challenges in bridging knowledge with real-world interactions and dynamics, particularly due to their limited understanding of physical commonsense (e.g. spatial, temporal, and fundamental physics principles). However, in practical applications, it is evident that focusing solely on perception or planning is insufficient; seamless collaboration requires a closed loop of perception, planning, and execution. Consequently, the concept of "grounding" is crucial, demanding precise alignment between model decisions and real-world physical details to ensure action-environment correspondence and prevent arbitrary reasoning. This capability is closely related to "**Physical Intelligence**", where agents generate intelligent behavior through physical interaction and environmental coupling. Compared to embodied and spatial intelligence, it places greater emphasis on the physical properties and dynamic constraints of the environment. As discussed in Section 5.1.2, physical attributes such as proprioceptive states are indispensable, with their importance increasingly recognized as critical in advancing trends. The startup Physical Intelligence aims to bring AI into the physical world. Its first large model, $\pi 0$ [150], has demonstrated general physical capabilities across multiple robots, showcasing the potential of physical intelligence in real-world applications. Physical Intelligence represents a development direction of AI rooted in physical laws, complementing embodied and spatial intelligence to drive AI systems toward a deeper understanding and manipulation of the physical world. This also highlights the core challenges and mission of future work, marking a key breakthrough toward truly intelligent collaborative systems.

7. Conclusions

This paper provides a comprehensive review of the integration of MLLMs with human–robot collaboration, summarizing the latest advancements, challenges, and potential research avenues. In particular,

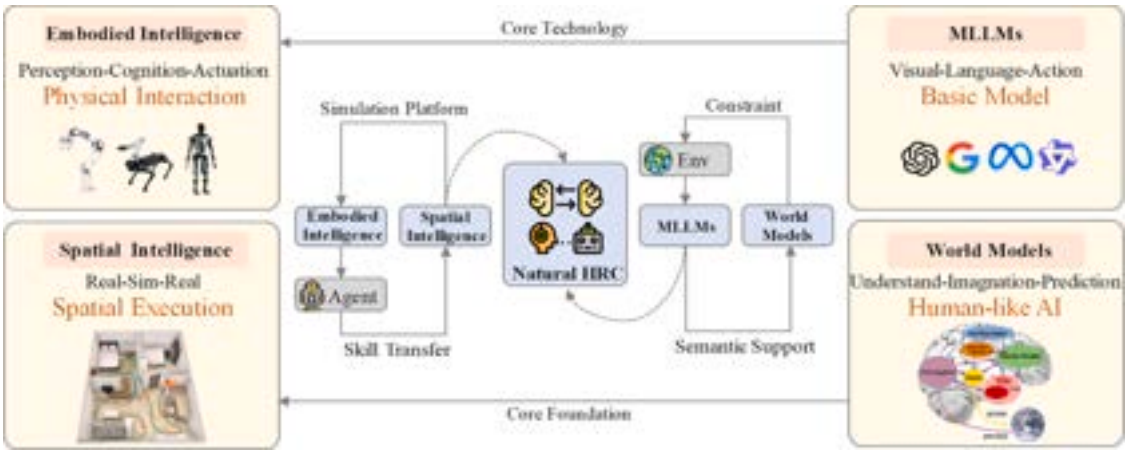

Fig. 11. A summary of the relationships between the key technologies involved in this work. They complement each other, moving toward human-like intelligence and natural HRC.

we emphasize the importance of the human-in-the-loop, considering aspects such as understanding human intentions, mutual cognitive recognition, and interaction between robots and AI agents. We explore cutting-edge algorithms such as VLA, diffusion models, and world models, outlining a pathway toward multimodal fusion and human-like intelligence. We also stress the significance of task-specific models and spatial skill learning, and explore opportunities to incorporate digital twins, augmented reality, and embodied intelligence simulation platforms in industrial and manufacturing contexts. Fig. 11 summarizes the key technologies. MLLMs and world models are core technologies for achieving embodied and spatial intelligence, respectively. Spatial intelligence provides virtual cyberspace for efficient, low-cost training and multi-skill learning. These complementary approaches collectively drive enhanced intelligence and natural interaction.

This work offers valuable insights for researchers in human-centered applications of MLLMs in HRC. However, facilitating deeper integration and deploying these studies into practical scenarios remains an urgent direction that warrants further exploration.

CRediT authorship contribution statement

Duidi Wu: Writing – original draft, Methodology, Conceptualization. **Pai Zheng:** Writing – review & editing, Supervision, Conceptualization. **Qianyou Zhao:** Validation, Investigation. **Shuo Zhang:** Validation, Investigation. **Jin Qi:** Supervision, Methodology, Funding acquisition. **Jie Hu:** Resources, Funding acquisition. **Guo-Niu Zhu:** Resources, Investigation. **Lihui Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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